

Sen Zhang

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Summary

Registered NY architect (NCARB) and computational design technologist, 9 years post-Master's. Sole developer behind Ennead's Design Technology stack: EnneadTab — a 450+ tool plugin platform with three U.S. Copyright registrations and 60 daily users across a 160-person firm — plus BimRunner, EnneadTabLLM, FieldDay, AIA-Sustainable-Reporting, and 6 more production tools. Independent work spans a real-estate networking platform (Realm) and agent-orchestrated AI operations across multiple live small-business ventures. Speaker: Advancing Computational Design 2026, Autodesk University 2024, Digital Built Week Americas 2023.

Experience

Ennead Architects LLP / Computational BIM Developer

New York, NY · 2022–Present

Architectural Designer, BIM Specialist

2020–2022

- **Computational BIM Developer — sole developer running the Design Technology department for a 160-person firm; full-spectrum role spanning platform development, BIM management, and complex-geometry triage**
 - + Architect and lead developer of EnneadTab — 450+ tools across Revit, Rhino, CAD, Adobe, Excel, Email, and LLM integration; three U.S. Copyright registrations (2024)
 - + BimRunner — Revit project-health analytics dashboard (Next.js 14 + Vercel Postgres); RESTful ingestion + dynamic scoring surfaces model debt before coordination crisis
 - + EnneadTabLLM — on-prem Ollama RAG monorepo for HIPAA / data-sovereign clients
 - + AIA-Sustainable-Reporting — CRON Excel pipeline + dashboard for AIA 2030 metrics; FieldDay — offline-first PWA for site inspection with PDF spatial pinning
 - + EnneaDuck — pixel-mascot community-engagement system that gamifies internal adoption of design-tech across 160 staff
 - + BIM management — Healthcare (NE US): NYU Langone, Northwell, Lenox Hill. Commercial (Asia): FTZ Shanghai, ByteDance Houhai HQ, Bilibili HQ
 - + Solution architect for large-model BIM data, complex families; in-house training and mentoring
 - + Rhino-to-Revit interoperability via zero-learning-curve internal tooling; triage of complex parametric and industrial-scale geometry
 - + Speaker: Autodesk University 2024 (San Diego), Advancing Computational Design Conference 2026 (Austin)
 - + Fluent in CPython, IronPython, Revit API, RhinoCommon API. Working proficiency in TypeScript / React / Next.js — shipped to production across 8+ personal and firm projects (LLM-assisted). LLM integration via Ollama, OpenAI, Anthropic, MCP

Arquitectonica / Designer, BIM Specialist, CG Visualization

New York, NY · 2017–2020

- 40 Hudson Yards (NYC office tower); Amazon HQ2 master planning (Chicago); 31 Parliament mid-rise residential tower (Toronto SmartCity)
- Curtis Block (Calgary); Warsaw Hotel (Poland); Newport Waterfront; Equinox Hotel (San Jose); Queens Sherbourne mixed-use with historical preservation (Toronto)

RPI Publication Dept. / Book-design, Asset Management

Troy, NY · 2016–2017

Plasma Studio / Summer Intern

Beijing, China · Summer 2015

Terreform ONE / Research, Design

New York, NY · Summer 2014

Ballistic Architecture Machine / Summer Intern

Beijing, China · Summer 2013

Selected Projects

Office Work — Ennead Architects

EnneadTab / Architect & Lead Developer

2021–Present

Firm-wide plugin ecosystem spanning Revit, Rhino, CAD, Adobe, Excel, Email, and on-prem LLM. Built and maintained solo. Designed end-user-first and adopting an invisible-design principle — silent updates, no install screen, no docs to read — which let it spread organically by word of mouth across the firm before any mandate, training rollout, or policy push. 500+ tools, three U.S. Copyright registrations (2024), 60 active daily users, 17×/person/workday, 63,627 logged events Q4 2025. Stack: Python (CPython + IronPython) + Revit/Rhino APIs.

EnneadTab Self-Healing Agent Ecosystem / Architect & Lead Developer

2025–Present

Autonomous agent layer wrapped around the EnneadTab-OS codebase that catches production errors, drafts fixes, and ships enhancements with minimal human intervention. Components: ErrorDump (live error capture from 60 daily users) → scheduled fix agent (auto-PR per fingerprint) → enhance agent (proactive code improvements). Featured in the Austin 2026 conference talk on agile tool introduction. Stack: Claude Code agents + GitHub Actions + Python + REST.

BimRunner / Architect & Full-Stack Developer

2026–Present

Production analytics dashboard that scores Revit project health and surfaces model debt before it becomes coordination crisis. Used by project leads across the firm to triage live work. Stack: Next.js + Vercel Postgres + Chart.js.

RenderPolisher / Architect & Developer

2025–2026

AI-driven render post-processing pipeline. Cleans and uplifts architectural visualizations through a configurable model chain; integrates into the Ennead delivery workflow via the EnneadTab toolbar. Stack: Next.js + Vercel Blob/KV + react-pdf book-flip viewer.

AIA-Sustainable-Reporting / Architect & Lead Developer

2026–Present

CRON Excel pipeline + frontend dashboard for AIA 2030 sustainable-design metrics. Single-source `.env.local` config across Next.js app, cron scripts, and CLI; Vercel CLI + Codespaces friendly. Closes the firm-wide reporting loop for the AIA 2030 commitment. Stack: Next.js + NextAuth + Drizzle ORM + Chart.js.

EnneadTabWiki / Architect & Lead Author

Ongoing

Public documentation portal for the EnneadTab tool family — auto-syncs tool catalog and usage data from the underlying registry, replacing a legacy hand-maintained static site. Stack: Next.js 16 + Drizzle + Vercel Postgres + Vercel Blob; semantic search via Gemini.

EmployeeData / Architect & Lead Developer

2025–2026

Decision-support system for Ennead partners on capital-equipment prioritization. Consolidated HR records (177 employees), weekly directory scrapes, and per-machine hardware reports into a single employee-by-machine view; surfaced that 70% of the 176-machine fleet was Retired and 81% of GPUs were 5+ years old, sequenced into a quarterly replacement plan through 2027 Q3. Replaced an ad-hoc Excel email workflow; gated to the partners and DT lead who approve capex. Delivered and retired April 2026 once the decision cycle completed. Stack: Python (Playwright scraper, psutil/WMI collector, FastAPI POST server, vector-store assistant) + GitHub Pages + Actions weekly cron.

EnneadTabLLM / Architect & Developer

2025–2026

On-premise AI assistant that lets architects ask plain-English questions across years of firm standards, templates, and project history — without sending any model data to an outside vendor. Solves the HIPAA / data-sovereign client problem and is the production basis of the Austin 2026 talk "Cloning the Expert." Stack: Ollama (on-prem) + Next.js proxy on Vercel.

FieldDay / Architect & Lead Developer

2026–Present

Offline-first PWA for site inspection on phones, tablets, and desktops. PDF-based spatial pinning, photos, voice notes, and comments with zero-connectivity capture and Vercel Pro cloud sync. Preserves spatial accuracy across drawing revisions. Stack: Next.js + NextAuth + Drizzle ORM + Vercel Blob, offline-first via service workers.

EnneadTabHome / Architect & Developer

2025–2026

Landing page and central hub for the EnneadTab application family at enneadtab.com. Next.js 14, Tailwind, mobile-friendly UI.

EnneaDuck / Designer & Developer

2022–Present

Community-mascot system across the EnneaTab ecosystem — pixel-art duck embedded throughout the toolset (achievements, status pings, gamified adoption hooks) plus a Next.js Duckland community page. Per user feedback, made the toolset feel more memorable and approachable — the warmth layer that helped a top-down rollout become a self-sustaining voluntary community.

2025 Performance Review / Architect & Lead Developer

2025–2026

Firm-wide employee-evaluation system. Excel ingest → interactive HTML report with software-proficiency tracking and visual analytics. Deployed to GitHub Pages.

Independent Projects — Outside Office

Realm / Digital Strategy Architect

2025–Present

Networking and event platform for architects, real-estate operators, and exhibition partners. Multi-step RSVP, admin/client dashboards, role-based partner gating, branded email flows. Stack: Next.js 16 + NextAuth + Drizzle + Vercel Postgres + Resend.

Atelier-T (Toni LLP) / Founder & Lead Engineer

2025–Present

In-development small venture exploring AI virtual hairstyle try-on for salons. Two prototypes: Toni (consumer webcam-to-hairstyle try-on) and Toni-Business (salon discovery + outreach drafting). Operating model uses a small agent team for routine ops while founder handles sales, demos, and decisions. Stack: Next.js 16 + Supabase + Gemini + Stripe + Resend.

Social-Media Ads Service / Lead Engineer

Ongoing

AI-workflow case study on live SMB ad operations — Claude-orchestrated agent team handles creative-brief intake, asset generation, scheduling, and performance reporting. Stack: Python monorepo (52 mp-* packages) — ComfyUI / RunningHub for image+video generation; mp-auto-generate orchestrator; mp-auto-post + mp-auto-engage + mp-engage-dashboard for posting and engagement loop.

Social-Media Channel — Android App / Lead Engineer

Ongoing

Native Android self-guidance app for social-media consumption — strips Reels/Shorts/algorithmic feeds from Instagram, YouTube, Facebook, and X via WebView CSS/JS injection, returning user agency over feed time. Stack: Kotlin 2.3 + Jetpack Compose + Room + Firebase Crashlytics. Pre-launch.

LakeHouse-Design.com / Architect & Developer

Ongoing

Rhino package of professional design tools — dynamic area takeoff, layer/material merging, object transformation, block management.

TimeBank LLP / Founder & Lead Engineer

2025–Present

Two-app side venture: TimeSheetHelper (gamified timesheet SaaS that turns time tracking into points and streaks for design teams) and TimeBank-Business (AI-first business ops hub that runs daily admin via 16 agent personas). Operating model: founder approves direction; agent team handles execution and reports daily. Stack: Next.js + Supabase + Claude Code agent commands; Bash-orchestrated multi-repo workspace.

> Note on visuals — the EnneaTab product family and the internal firm tools above are Ennea IP and cannot be shown in a public résumé. A desensitized live walkthrough is available on request.

Speaking & Recognition

Advancing Computational Design Conference / Speaker

Austin, TX · 2026

The Design of Design — Enabling Agile Introduction of New Tools to Scale Design Efficiency

Autodesk University / Session Speaker

San Diego, CA · 2024

Revit As A Game Engine

Digital Built Week Americas / Panel Speaker

Dallas, TX · 2023

Promoting Computational Design to Non-Programmers

AEC Hackathon / Team — Honorable Mention

New York, NY · 2023

Educational Tool Connecting Junior and Senior Students

Credentials & Technical Stack

- Registered Architect, New York (NCARB)
- Languages: Python (CPython, IronPython) — primary; TypeScript / React / Next.js — applied (LLM-assisted)
- Tooling: Revit API, RhinoCommon API, Grasshopper, Dynamo, MCP, LLM integration (Ollama, OpenAI, Anthropic)
- Spoken: English, Mandarin Chinese (native)

Education

Columbia University GSAPP / M.S. Advanced Architecture Design	New York, NY · 2016–2017
Rensselaer Polytechnic Institute / Bachelor of Architecture	Troy, NY · 2012–2016
Tongji University / Major in Urban Planning	Shanghai · 2010–2012

Awards & Recognition

• EnneadTab — U.S. Copyright Office — 3 registrations, authorship credited to Sen Zhang	2024
• Hyperloop Campus, Las Vegas — Honorable Mention	2020
• Greenway to Ideal City, Chengdu — 2nd Place	2019
• Tokyo Anti-Library — Honorable Mention	2018
• Non Architecture: Dance — Finalist	2018
• AIA Henry Adams Award	2016
• Magna Cum Laude Honors — Dean's List 2012–2016	2016
• George T. Droste Scholarship — Structural engineering study	2015
• Faculty Selection Award — Academic proficiency in Architecture	2015